Series_16-18_21-33

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Chapter 1

File Index

1.1 File List

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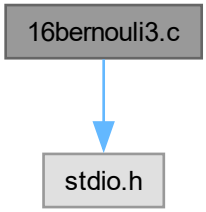
Chapter 2

File Documentation

2.1 16bernouli3.c File Reference

`#include <stdio.h>`
Include dependency graph for 16bernouli3.c:



Functions

- int main ()

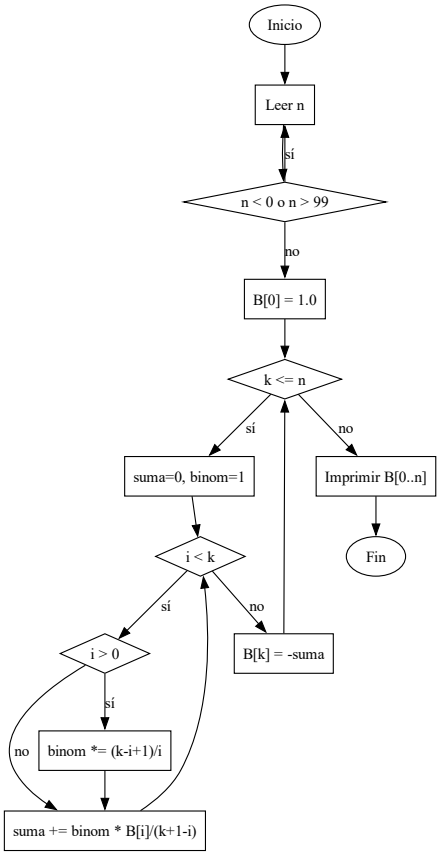
  Lee n y calcula los números de Bernoulli B[0..n].

2.1.1 Function Documentation

2.1.1.1 main()

`int main ()`

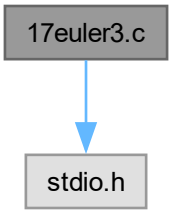
Lee n y calcula los números de Bernoulli B[0..n].



2.2 17euler3.c File Reference

`#include <stdio.h>`

Include dependency graph for 17euler3.c:



Macros

- #define PI 3.141592653589793

Functions

- int main (int argc, char ∗argv[ ])

  Lee k e iter y calcula E_k mediante una serie alternante.

2.2.1 Macro Definition Documentation

2.2.1.1 PI

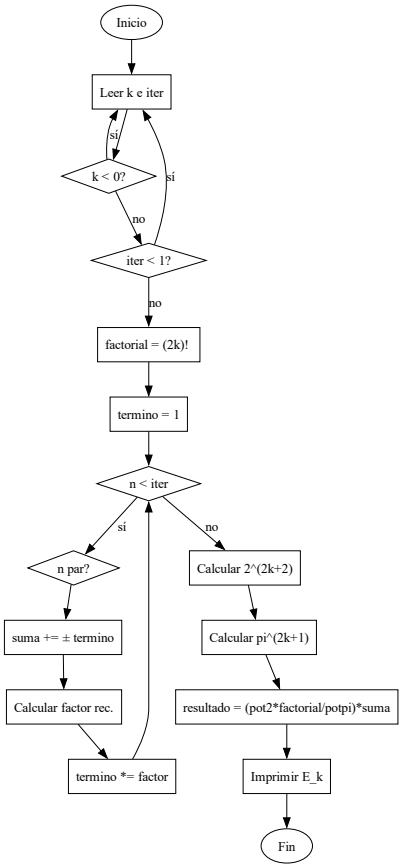
```
#define PI 3.141592653589793
```

2.2.2 Function Documentation

2.2.2.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

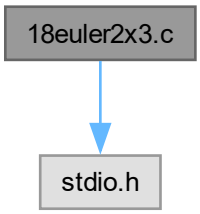
Lee k e iter y calcula E_k mediante una serie alternante.



2.3 18euler2x3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 18euler2x3.c:



Functions

- int main ()

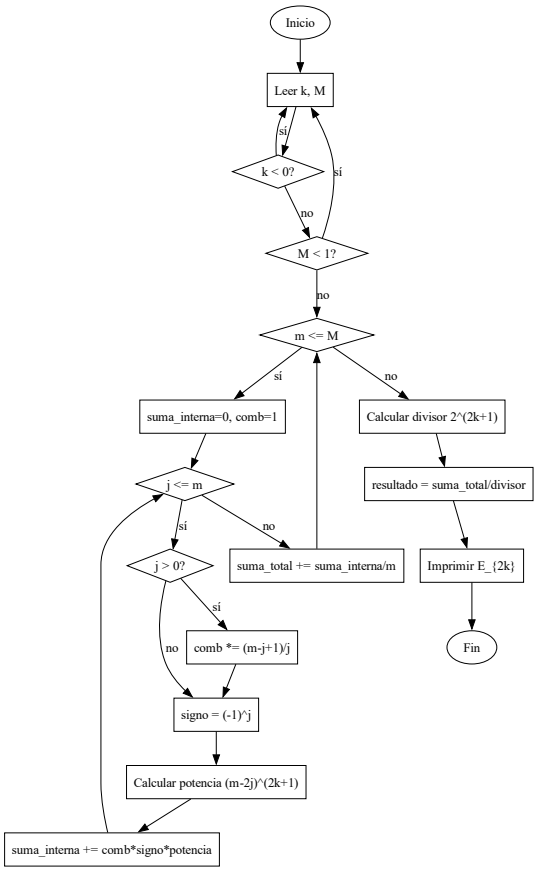
  Lee k y M para aproximar E_{2k} con una doble sumatoria.

2.3.1 Function Documentation

2.3.1.1 main()

```
int main ()
```

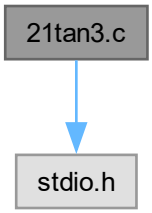
Lee k y M para aproximar E_{2k} con una doble sumatoria.



2.4 21tan3.c File Reference

`#include <stdio.h>`

Include dependency graph for 21tan3.c:



Macros

- #define PI 3.141592653589793

Functions

- int main (int argc, char ∗argv[ ])

  Lee x y n, convierte grados a radianes y calcula tan(x) por serie.

2.4.1 Macro Definition Documentation

2.4.1.1 PI

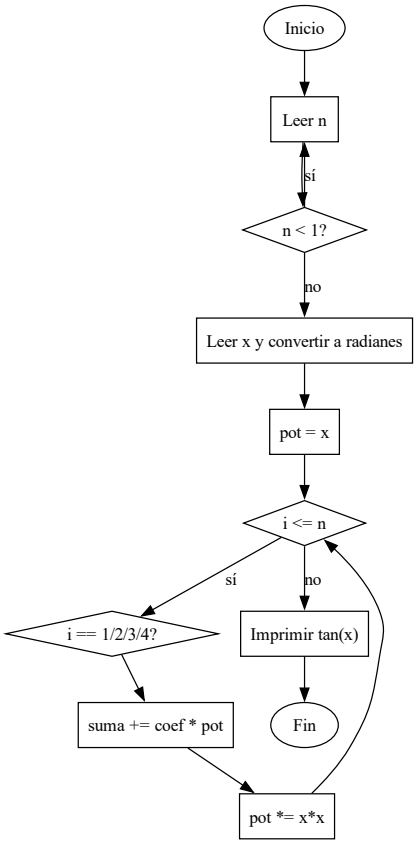
```
#define PI 3.141592653589793
```

2.4.2 Function Documentation

2.4.2.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

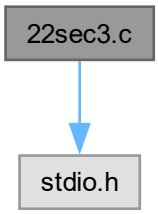
Lee x y n, convierte grados a radianes y calcula tan(x) por serie.



2.5 22sec3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 22sec3.c:



Macros

- #define PI 3.141592653589793

Functions

- int main (int argc, char *argv[])

  Lee x y n para aproximar sec(x) con una serie de Maclaurin.

2.5.1 Macro Definition Documentation

2.5.1.1 PI

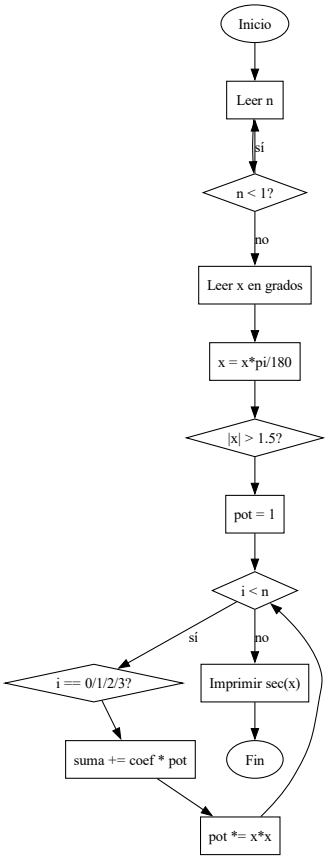
```
#define PI 3.141592653589793
```

2.5.2 Function Documentation

2.5.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

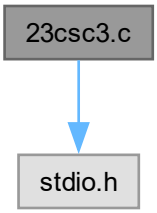
Lee x y n para aproximar sec(x) con una serie de Maclaurin.



2.6 23csc3.c File Reference

`#include <stdio.h>`

Include dependency graph for 23csc3.c:



Macros

- #define PI 3.141592653589793

Functions

- int main (int argc, char ∗argv[ ])

  Lee n y x para aproximar csc(x) por la serie de Maclaurin.

2.6.1 Macro Definition Documentation

2.6.1.1 PI

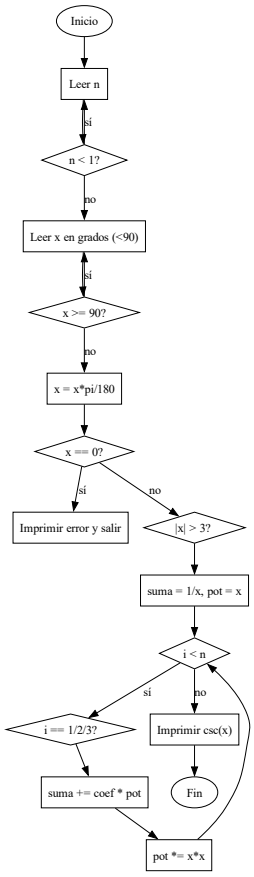
```
#define PI 3.141592653589793
```

2.6.2 Function Documentation

2.6.2.1 main()

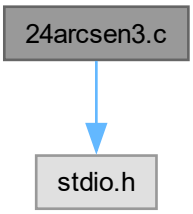
```
int main (
            int argc,
            char * argv[ ])
```

Lee n y x para aproximar csc(x) por la serie de Maclaurin.



2.7 24arcsen3.c File Reference

`#include <stdio.h>`
Include dependency graph for 24arcsen3.c:



Functions

- int main (int argc, char ∗argv[ ])

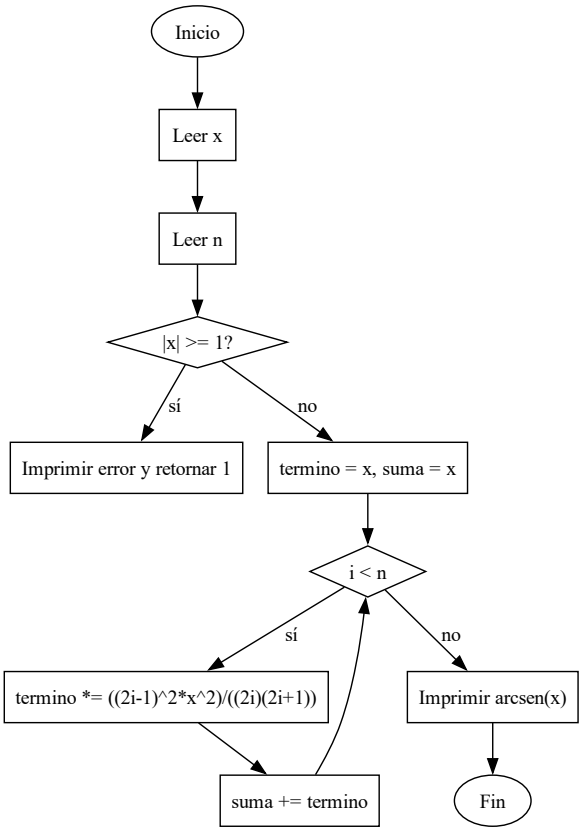
  Lee x y n y aproxima arcsen(x) mediante la serie de Maclaurin.

2.7.1 Function Documentation

2.7.1.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

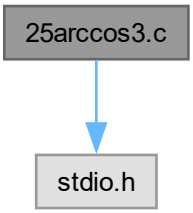
Lee x y n y aproxima arcsen(x) mediante la serie de Maclaurin.



2.8 25arccos3.c File Reference

`#include <stdio.h>`

Include dependency graph for 25arccos3.c:



Macros

- #define PI 3.141592653589793

Functions

- int main (int argc, char ∗argv[ ])

  Lee x y n y aproxima arccos(x) usando arccos(x)=pi/2 - arcsen(x).

2.8.1 Macro Definition Documentation

2.8.1.1 PI

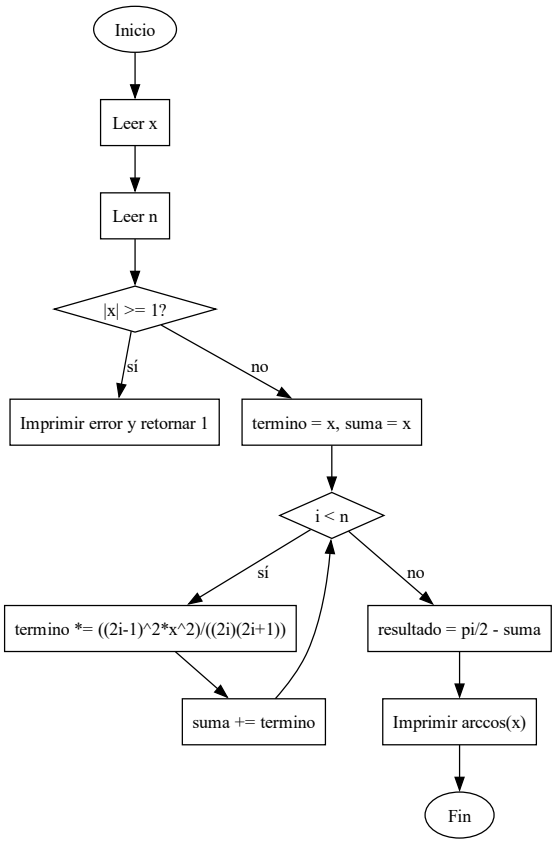
```
#define PI 3.141592653589793
```

2.8.2 Function Documentation

2.8.2.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

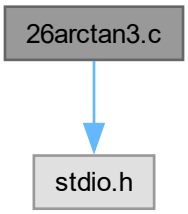
Lee x y n y aproxima arccos(x) usando arccos(x)=pi/2 - arcsen(x).



2.9 26arctan3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 26arctan3.c:



Macros

- #define PI 3.141592653589793

Functions

- int main (int argc, char ∗argv[ ])

  Lee x y n y aproxima arctan(x) con ajuste de dominio cuando |x|>=1.

2.9.1 Macro Definition Documentation

2.9.1.1 PI

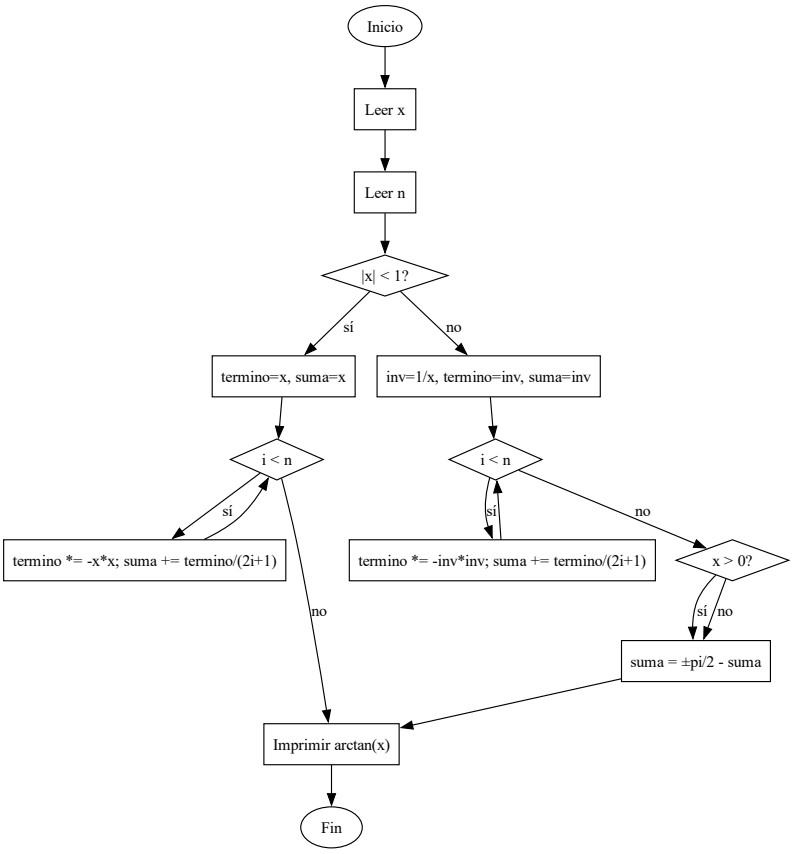
```
#define PI 3.141592653589793
```

2.9.2 Function Documentation

2.9.2.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

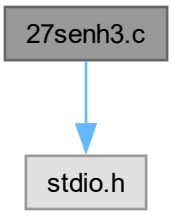
Lee x y n y aproxima arctan(x) con ajuste de dominio cuando |x|>=1.



2.10 27senh3.c File Reference

`#include <stdio.h>`

Include dependency graph for 27senh3.c:



Functions

- int main (int argc, char *argv[ ])

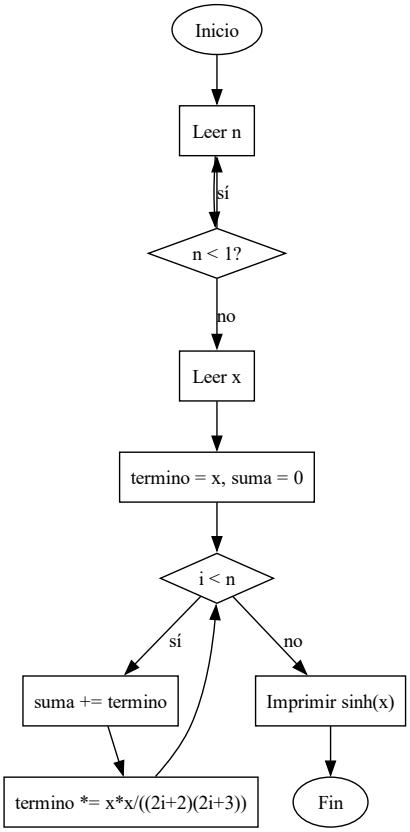
  Lee x y n y aproxima sinh(x) por la serie de Taylor.

2.10.1 Function Documentation

2.10.1.1 main()

```
int main (
            int argc,
            char * argv[])
```

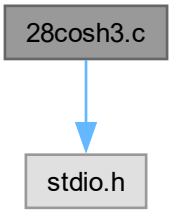
Lee x y n y aproxima sinh(x) por la serie de Taylor.



2.11 28cosh3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 28cosh3.c:



Functions

- int main (int argc, char ∗argv[ ])

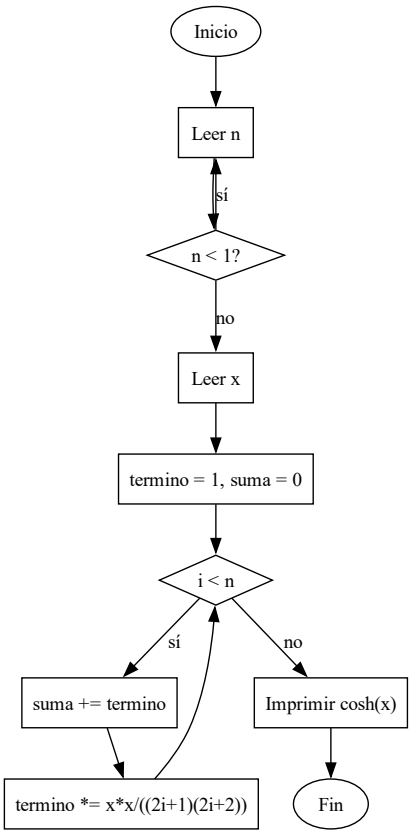
  Lee x y n y aproxima cosh(x) por la serie de Taylor.

2.11.1 Function Documentation

2.11.1.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

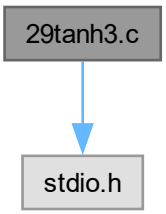
Lee x y n y aproxima cosh(x) por la serie de Taylor.



2.12 29tanh3.c File Reference

`#include <stdio.h>`

Include dependency graph for 29tanh3.c:



Functions

- int main ()

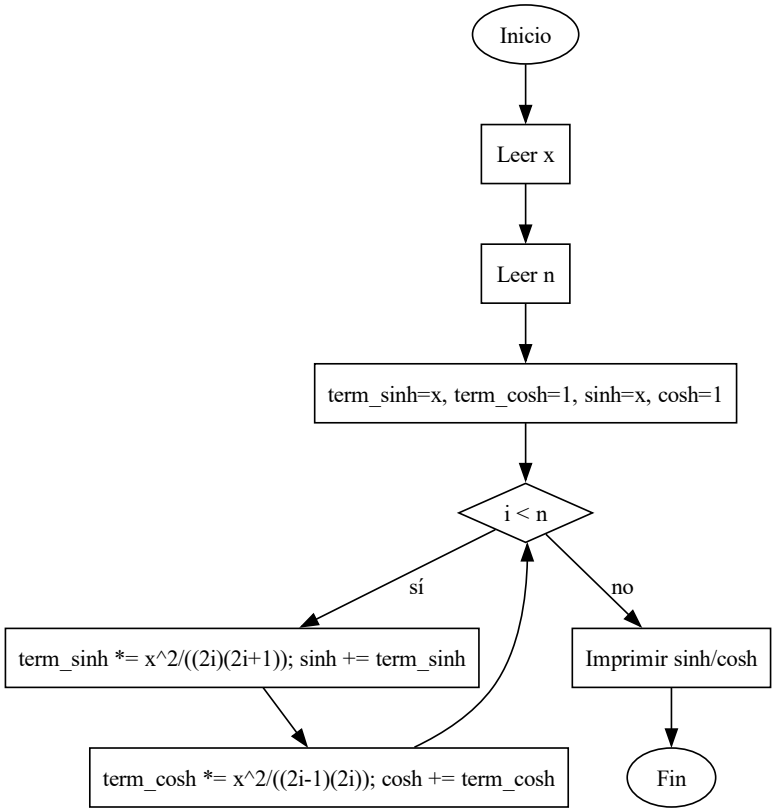
  Lee x y n y calcula tanh(x) como sinh(x)/cosh(x) por series.

2.12.1 Function Documentation

2.12.1.1 main()

```
int main ()
```

Lee x y n y calcula tanh(x) como sinh(x)/cosh(x) por series.



Inicializacion de los primeros terminos

Calculo de la serie

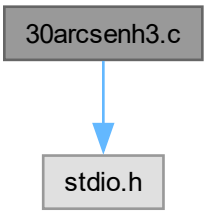
Relacion recursiva para sinh

Relacion recursiva para cosh

2.13 30arcsenh3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 30arcsenh3.c:



Functions

- int main ()

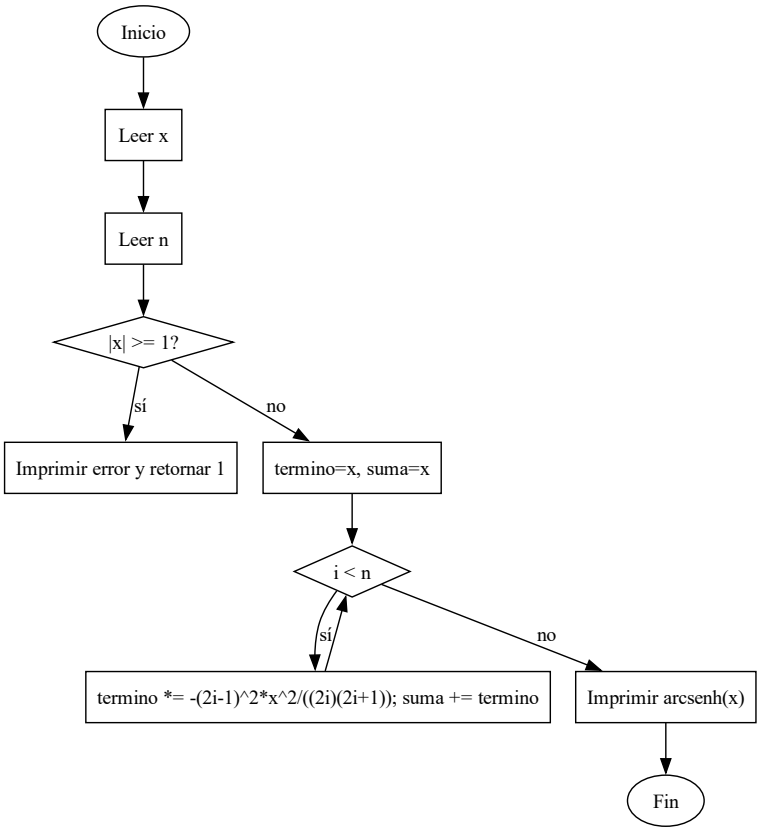
  Lee x y n y aproxima arcsenh(x) mediante serie de Taylor con signo alternante.

2.13.1 Function Documentation

2.13.1.1 main()

```
int main ()
```

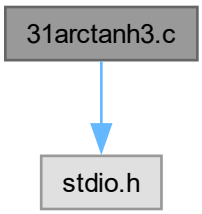
Lee x y n y aproxima arcsenh(x) mediante serie de Taylor con signo alternante.



2.14 31arctanh3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 31arctanh3.c:



Functions

- int main ()

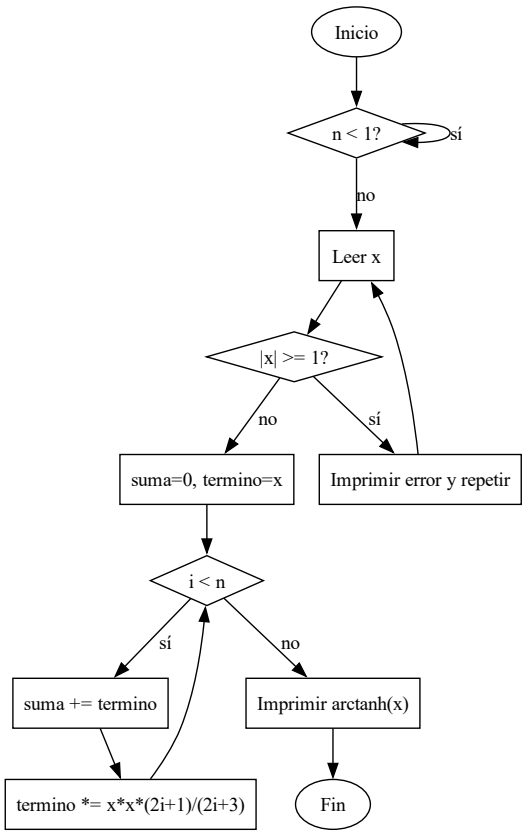
  Lee n y x, valida dominio y aproxima arctanh(x) con serie de Maclaurin.

2.14.1 Function Documentation

2.14.1.1 main()

```
int main ()
```

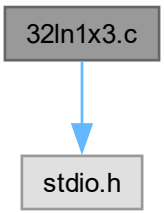
Lee n y x, valida dominio y aproxima arctanh(x) con serie de Maclaurin.



2.15 32ln1x3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 32ln1x3.c:



Functions

- int main (int argc, char ∗argv[ ])

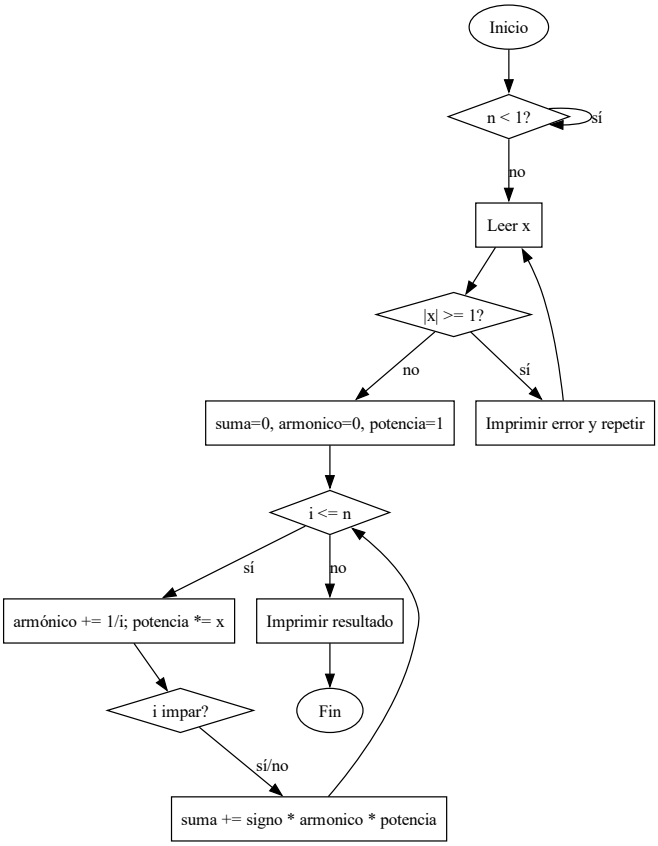
  Lee n y x, valida el dominio y aproxima ln(1+x)/(1+x) por serie.

2.15.1 Function Documentation

2.15.1.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

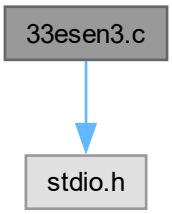
Lee n y x, valida el dominio y aproxima ln(1+x)/(1+x) por serie.



2.16 33esen3.c File Reference

```
#include <stdio.h>
```

Include dependency graph for 33esen3.c:



Functions

- int main (int argc, char ∗argv[ ])

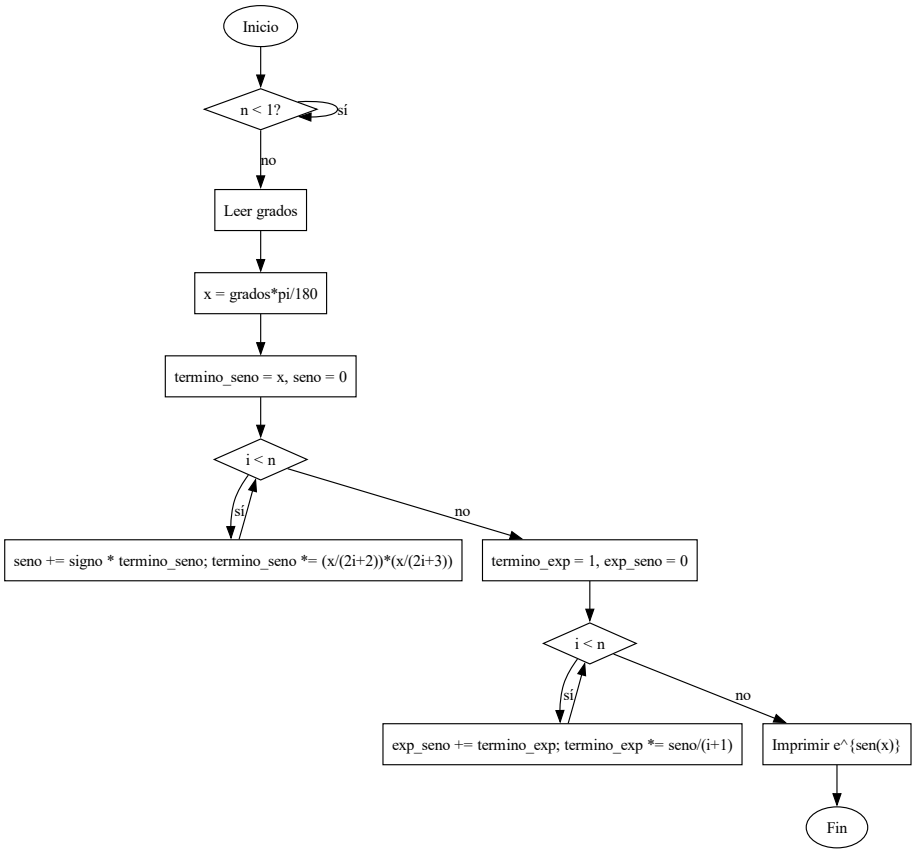
  Lee n y grados y calcula e^{sen(x)} mediante series optimizadas.

2.16.1 Function Documentation

2.16.1.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

Lee n y grados y calcula e^{sen(x)} mediante series optimizadas.


Inicio
n < 1?
sí
no
Leer grados
x = grados*pi/180
termino_seno = x, seno = 0
i < n
sí
no
seno += signo * termino_seno; termino_seno *= (x/(2i+2))*(x/(2i+3))
termino_exp = 1, exp_seno = 0
i < n
sí
no
exp_seno += termino_exp; termino_exp *= seno/(i+1)
Imprimir e^{sen(x)}
Fin

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